UNDERBOT / THE WHOLE CONNECTION MAP

Team presentation • Xiaomi T12 + Kinect v1 (assumed) + two IMUs + dual arms • Design proposal 0.2

Read top to bottom. This explains the system; connector pin numbers and several hardware choices still need verification.

Orange = power (+ and return) | Blue = data / commands | Red = hardware stop | Dashed purple = unresolved / future

1 WHERE THE ENERGY COMES FROM

The robot battery and the charging-station battery are separate. Each has its own matching battery protection and charging settings.

Station solar panels
2 × 100 W in series
Outdoor energy source

→ **Station solar controller**
Victron MPPT candidate
PV fuse / DC disconnect

→ **Station buffer battery**
8S LiFePO4 + BMS + fuse
25.6 V / 50 Ah nominal

→ **Robot charging unit**
True buck-boost Li-ion charger
25.2 V / 4 A candidate

→ **Protected docking contacts**
Fuse, current limit, polarity check
Dock detect → prevent motion

Optional roof panel
5–6 W peak
Supplemental energy only

→ **Robot solar charger**
Boost MPPT for 6S Li-ion
Not a direct panel-to-pack wire

Mains fallback charger
Dedicated 6S Li-ion charger
25.2 V / 4 A candidate

→ **Charge-source manager**
Select / safely manage sources
No raw parallel charger outputs

→ **ROBOT PACK: 6S3P + BMS**
18 matched 21700 cells
25.2 V full; ~266 Wh plan
Cell sense wires + temperature probe

P0 = protected battery output → section 2

BMS = battery protection, not a charger. All charging returns must use the correct BMS port. T12 charging remains separate below.

2 HOW POWER REACHES EACH PART

A buck converter reduces voltage. One shared voltage supply can feed several compatible devices; every branch needs appropriate protection.

BATTERY BUS

P0 → Main battery fuse
Mounted close to the pack; rating TBD

Main isolator / keyed switch
Manual service power disconnect

Current and voltage monitor
Rated external shunt + INA228
Measures robot load → I2C to ESP32
Bus surge / reverse protection: TBD

Every power cable needs a return
GND = protected electrical return.
Motor current must not flow through the Pi or sensor ground wires.

E-stop → reset / watchdog circuit
Removes the hardware motion permit

P1 Fuse → compute buck → USB-C source
5.1 V target; verify power at Pi under load
→ **Raspberry Pi 5 + its peripherals**
USB devices / SSD / fan included in budget

P2 Fuse → 12 V sensor buck
Confirm Kinect adapter voltage and polarity
→ **Kinect v1 power / USB adapter**
Adapter supplies power; USB carries data to Pi

P3 Fuse → 5 V auxiliary buck
Independent of Pi supply; common system return
→ **Audio / status LEDs / accessories**
Each module must accept this voltage

P4 P3 → 3.3 V logic regulator
Add module decoupling; verify breakout ratings
→ **Body IMU + head IMU + sensor logic**
I2C mux / ToF / INA228 logic as selected

P5 Fuse → actuator buck → power bus
Voltage and current depend on selected servos
→ **Arms + grippers + camera pan/tilt**
Separate power wires; signals come from section 3

Normally-open actuator contactor
Heavy power switch; opens when permit is lost
Coil driver + suppression + precharge interlock: design TBD

E-stop target: stop motion; keep Pi / Kinect alive.
Arm brakes / gravity behavior must be designed too.

Not shown as ready-to-fit parts: transient protection, precharge/inrush limit, coil suppression, fuse values, cable gauges and connectors.

3 HOW THE ROBOT TALKS AND SENSES

Power labels P1–P5 refer to section 2. Data connections do not replace a device's power supply. The ESP32 handles low-level control.

Raspberry Pi 5 [P1]
ROS / perception / navigation
Network ↔ team computer
USB ↔ Kinect v1 [P2]

↔ **ESP32 controller [Pi USB power]**
Commands + feedback over USB
No simultaneous external 5 V feed
Heartbeat → hardware watchdog

↔ **I2C sensor bus / mux [P4]**
Routes one short bus into separate sensor channels
Avoids duplicate addresses

↔ **Two separate IMUs + optional ToF x4 [P4]**
Channel 0: body IMU fixed to chassis
Channel 1: head IMU moves with camera
Other channels: front / rear / left / right ToF

Pi accessory connections
SSD → USB or chosen PCIe HAT
Mic / audio interface → Pi
Pi audio → amplifier → speaker
Fan → correct fan connection
Optional AI board: verify Pi interface

Power / temperature feedback
INA228 → ESP32 I2C bus
BMS serial → voltage-safe adapter
Second NTC → divider / ADC
Pi has no general analog input

Actuator signal interface
PWM servos: PCA9685 [P4]
Smart servos: matching bus
Select one suitable approach
Output defaults disabled

→ **Left arm / right arm / grippers / pan-tilt**
Signal wires from the selected driver → actuators
Motor power from P5 → separate distribution
Servo voltage, stall current and pinouts still TBD
Camera IMU is a sensor, not a pan/tilt controller

I2C = short sensor-data bus. Use correct pull-ups and local capacitors; confirm interrupt/reset wires and cable lengths after module selection.

4 THE XIAOMI BASE: KEEP ITS POWER SEPARATE FOR THE FIRST PROTOTYPE

This is a differential-drive vacuum base, not mecanum. Do not assume the existing controller exposes an open motor or LiDAR interface.

Original T12 battery / dock
14.8 V-rated base system
20 V-rated charging input
Never parallel with the new 6S pack

→ **T12 original controller**
Own verified hardware stop required
Power / commands → wheel motors
Wheel encoders → feedback

↔ **Existing base sensors**
LiDAR + bump + cliff sensors
OEM gyro is an additional sensor
Their internal pinouts are unverified

⇢ **Pi / ESP32 integration — TBD**
Model-specific data / control adapter
Motor commands + odometry + LiDAR
Discover protocol before wiring

Future single-pack option: 6S → separately fused, validated base supply + motion disconnect → T12. Do not build from voltage alone.

Check battery-sense pins, peak motor current and braking energy first. Opening the new pack contactor does NOT stop an OEM-battery-powered base.

5 WHAT THE TEAM SHOULD UNDERSTAND BEFORE BUILDING

A custom PCB is optional for the prototype. Start with protected modules and a labelled wiring harness; consolidate after interfaces are proven.

The small parts do important jobs
Fuse: protects a wire branch against excess current.
Buck: reduces voltage; it does not create extra energy.
Contactor: electrically operated high-current switch.
Shunt: small resistor used to measure current.
BMS: monitors/protects cells and pack temperature.

Important design boundaries
USB-C source must support the Pi's power needs.
Do not power actuators through a sensor board.
No raw 6S into T12, Pi, ESP32 or ordinary servos.
Six 2.7 V supercaps are NOT suitable across 6S.
Standalone emergency beacon needs its own protected battery/charger; it is optional, not a shared power rail.

Still needed to finish the wiring
Exact Kinect / adapter and T12 model labels.
Arm / gripper / pan-tilt actuator models and counts.
IMU / ToF boards, audio, storage and cooling choices.
Peak-current measurements and base API discovery.
Then: actual pin numbers, fuses, wires and PCB layout.

Runtime is a target, not a measurement: 266 Wh × 80% usable × 87% efficiency gives ~6.6 h at 28 W, or ~3.1 h at 60 W.

Based on the user's corrected component list and manufacturer references in README.md. Solid lines describe proposed routes, not bench-tested wiring.